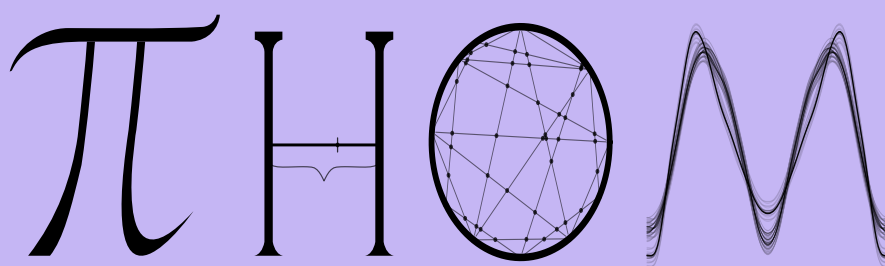


CANONICAL GEOMETRY · VERSION 1.1

THOM Typography Specification

Construction, optical alignment, measurement, and SVG
→ Three.js translation



460 × 120 master · 89u cap height · refined audit geometry

Status and authority

This specification makes the refined THOM artwork canonical for **shape, proportion, and placement**. The canonical wordmark is [thom-canonical.svg](#); the machine-readable contract is [thom-typography-metrics.json](#).

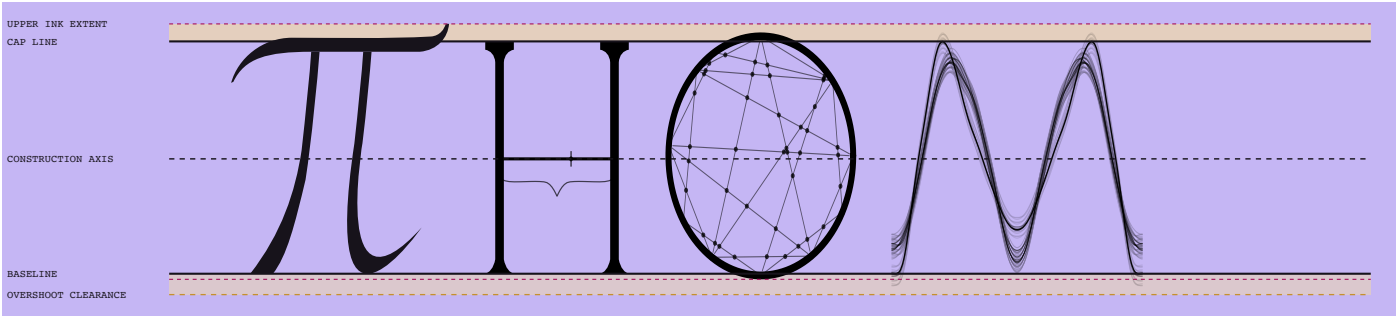
The wordmark occupies a **460 × 120 master-unit artboard**. The former 460 × 152 audit board is not a second geometry system: its lower 32 units are annotation space. One master unit (1u) is one SVG user unit.

Typographic rule. The baseline is the nominal alignment datum, not necessarily the lowest visible pixel. Filled contours, stroked centerlines, and round caps must be measured separately.

Evidence rule. Geometry and accessibility constraints eliminate known defects; they do not prove universal beauty. The H's golden-ratio division is an identity narrative and testable design prior, not a psychophysical optimum.

Authority	Canonical source	Rule
Shape and placement	<code>docs/brand/typography/thom-canonical.svg</code>	Governs glyph outlines, construction, relative scale, and spacing.
Numeric geometry	<code>docs/brand/typography/thom-typography-metrics.json</code>	Governs lines, frames, bounds, overshoot, grid coordinates, and conversion constants.
Color and materials	<code>src/brand/thom/geometry.ts</code>	Preserve the existing palette, metallic ramps, glow, stroke stacks, and source-energy compensation.
Motion and WebGL behavior	<code>src/brand/thom/threeScene.ts</code> and generated brand data	Preserve timing, reveal order, animated construction, and material behavior during migration.
Optical profiles	<code>src/brand/thom/opticalProfile.ts</code> and <code>src/brand/thom/svg.ts</code>	Select display, compact, or micro detail from rendered size while preserving the canonical silhouettes.

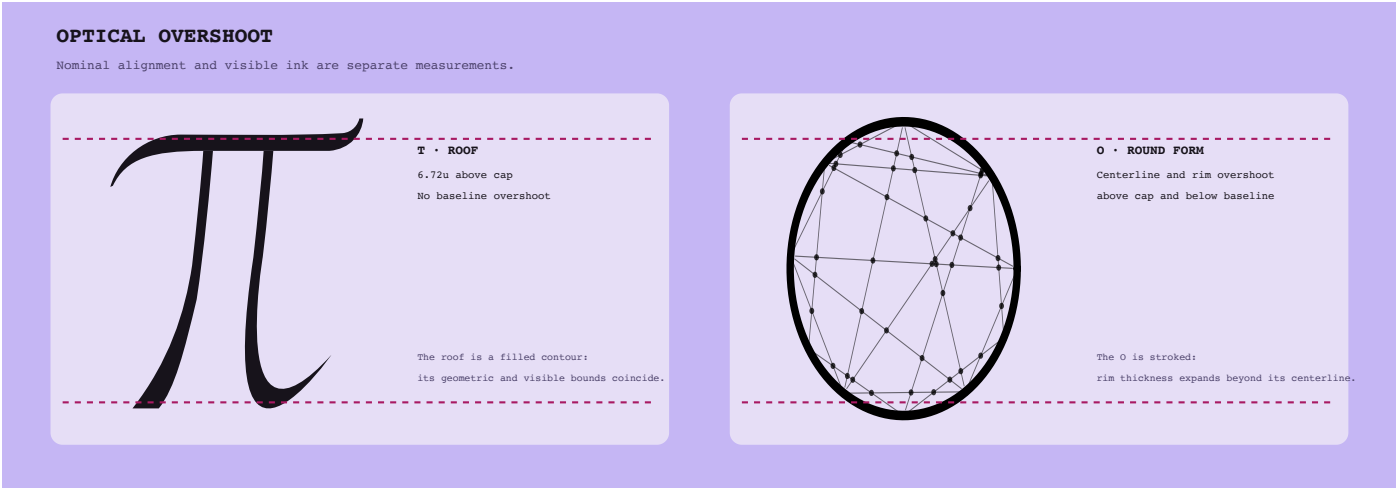
Vertical metrics



Metric	SVG y	Three.js y	Definition
Upper ink extent	8.30	111.70	Highest rendered ink, set by the T roof. It is an observed ink bound, not a font-wide alignment line.
Cap line	15.00	105.00	Nominal top alignment for H and the principal M waveform.
Horizontal construction axis	60.00	60.00	Shared geometric axis through the H proportion split and O construction.
Baseline	104.00	16.00	Nominal lower alignment for filled T/H contours and the principal M waveform.
Lowest visible ink	106.15	13.85	Rendered bottom of the O perimeter at the specified audit weight.
Lower overshoot clearance	112.00	8.00	Reserved clearance/safety boundary; it is not an ink target.

The cap height is **89u**. At the reference width of 2300 px, the export scale is **5 px/u**, so the cap height renders at 445 px.

Optical compensation



- **T**: the roof reaches $y = 8.2755$, or $6.7245u$ above the cap line. Its pillar terminals land on the baseline with no lower ink overshoot.
- **O**: the perimeter centerline extends beyond both cap line and baseline; the visible rim extends farther by half the transformed stroke weight. This is conventional round-form compensation, not misalignment.
- **H**: the filled terminals land on the cap line and baseline. Its stem width and terminal/serif width are distinct metrics.
- **M**: the principal centerline aligns to cap line and baseline; round strokes and layered fine strands add a small visible-ink extension.

Glyph metric sheets

Bounds are listed as **left, top, right, bottom** in master coordinates. “Geometric” uses SVG vector bounds; “visible ink” is verified at 20 px/u with alpha ≥ 0.5 and is reported to 0.05u precision.

Glyph	Design frame	Geometric bounds	Visible-ink bounds	Upper ink overshoot	Lower ink overshoot
T	20–106	23.72, 8.28, 107.14, 104.00	23.75, 8.30, 107.15, 104.00	6.70	0.00
H	113.5–182.5	120.70, 15.00, 176.25, 104.00	120.83, 15.00, 176.13, 104.00	0.00	0.00
O	187.875–264.875	190.26, 11.74, 262.87, 105.98	190.00, 11.60, 263.00, 106.15	3.40	2.15
M	275–396	276.60, 11.77, 372.60, 108.61	276.35, 14.70, 372.85, 104.70	0.30	0.70

T — three independent filled contours

The T is built from **top-bar**, **left-pillar**, and **right-pillar** contours. Each contour retains paired outer/inner Bézier edges. The canonical transform is `translate(22, -0.222) scale(0.86, 1.03)`. Curve edits must move corresponding anchors and adjacent handles along the local normal; never replace the contours with simplified polygons or non-uniformly scale the whole letter.

H — stem and terminal measures

- Construction-axis stem width: **3.3004u**.
- Cap/baseline terminal width: **11.5440u**.
- Local pillar centers after the audit adjustment: **28u** and **72u**.

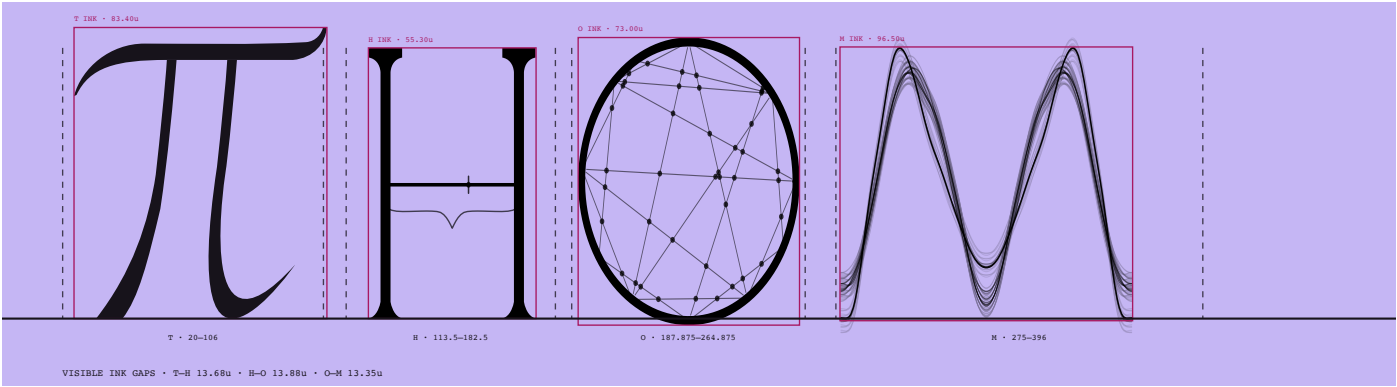
O — centerline versus rim

- Side perimeter weight: **2.2994u**.
- Cap/baseline perimeter weight: **2.9788u**.
- Perimeter centerline bounds, local: **8.652, 13.058, 79.358, 104.662**.
- Node/chord construction remains integral to the canonical O; the outer circle is not a substitute for the internal network.

M — layered waveform

The M remains a textural Fourier construction. Four fine-strand copies are added at local y offsets **−2.4, −1.2, +1.2, +2.4**, each at 0.4 opacity. Existing strand widths are unchanged.

Horizontal placement and spacing



The design frame is the nominal placement/advance region. The ink box is the actual rendered silhouette. Sidebearings are therefore measured from the frame edge to the visible ink—not inferred from a character's nominal 100u source cell.

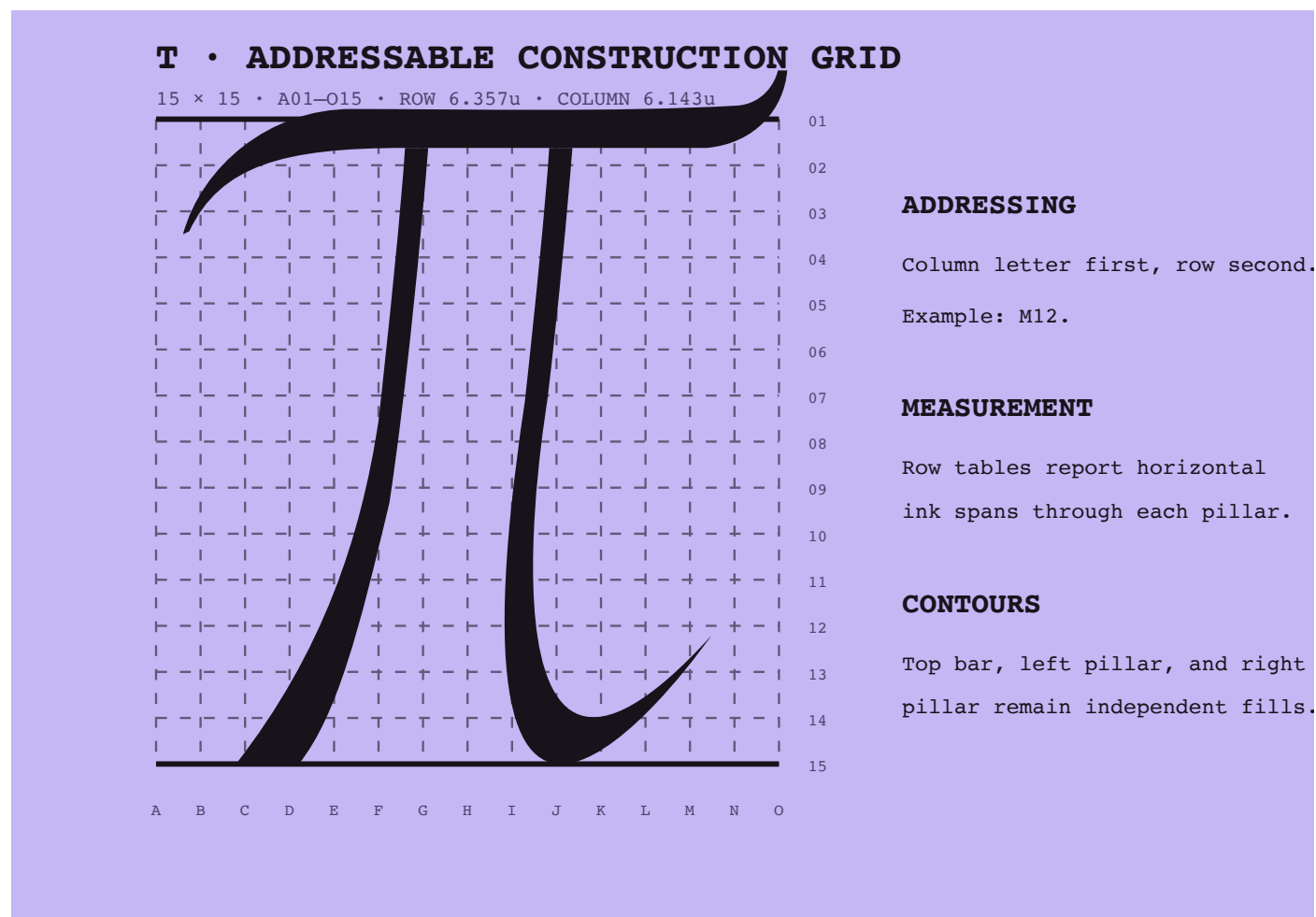
Pair	Visible-ink gap
T-H	13.6750u
H-O	13.8750u
O-M	13.3500u

The three gaps remain optically—not mechanically—defined, but now cluster within **0.5250u**. Their near-equality is a calibrated prior to be validated in use, not a claim that equal spacing is universally preferred.

Placement transforms

Glyph	Master transform	Design frame
T	$\text{translate}(22, -0.222) \cdot \text{scale}(0.86, 1.03)$	20–106
H	$\text{translate}(98.475, 0); \text{local pillar scale } x = 0.74$	113.5–182.5
O	$\text{translate}(182.5, -8.4) \cdot \text{scale}(0.88, 1.14)$	187.875–264.875
M	$\text{translate}(274.6, -26.7) \cdot \text{scale}(1, 1.49)$	275–396

T construction grid



Grid addresses use the **column letter first** and the **two-digit row second**: **A01** through **O15**. Rows are equally spaced between cap line and baseline; columns are equally spaced across the T design frame.

- Row spacing: **6.357143u**.
- Column spacing: **6.142857u**.
- Horizontal measurements report contiguous filled-ink spans through each independent pillar contour.
- Multiple values indicate distinct ink runs where a terminal or concavity crosses the same guide.

T pillar widths by row

Row	SVG y	Left pillar	Right pillar
01		15.0000	—
02		21.3571	3.174u
03		27.7143	3.153u
04		34.0714	3.155u
05		40.4286	3.154u
06		46.7857	3.138u
07		53.1429	3.094u
08		59.5000	3.148u
09		65.8571	3.549u
10		72.2143	3.877u
11		78.5714	4.411u
12		84.9286	5.328u
13		91.2857	6.560u
14		97.6429	7.903u
15		104.0000	—

T vertical intersections

Column	SVG x	Top bar	Left pillar	Right pillar
A	20.0000	—	—	—
B	26.1429	2.444u	—	—
C	32.2857	3.840u	1.649u	—
D	38.4286	5.109u	10.964u	—
E	44.5714	5.635u	14.855u	—
F	50.7143	5.407u	17.619u	—
G	56.8571	5.332u	8.395u	—
H	63.0000	5.292u	—	—
I	69.1429	5.262u	—	28.418u
J	75.2857	5.256u	—	23.565u + 9.444u
K	81.4286	5.291u	—	4.639u
L	87.5714	5.383u	—	2.431u
M	93.7143	5.551u	—	0.811u
N	99.8571	4.925u	—	—
O	106.0000	4.167u	—	—

The roof’s vertical span is deliberately non-uniform: its long sweep is thinner through the center and resolves into sharpened terminals. Pillar readings become discontinuous near the feet because the horizontal sample can cross a terminal turn more than once.

Scaling and export

Master units

Keep geometry in the 460 × 120 master whenever possible. Scale only at the final presentation boundary.

For a target width (W):

$$\text{scale} = W / 460$$

At the reference export, (W = 2300), so $\text{scale} = 5 \text{ px/u}$. The corresponding height is $120 \times 5 = 600 \text{ px}$.

For a constrained viewport, use a uniform contain scale:

```
const scale = Math.min(viewportWidth / 460, viewportHeight / 120);
const offsetX = (viewportWidth - 460 * scale) / 2;
const offsetY = (viewportHeight - 120 * scale) / 2;
```

Never fit the complete wordmark with independent X and Y scales. Glyph-specific transforms already belong to the canonical construction and must not be reinterpreted as responsive distortion.

Optical profiles

Profile selection uses the **rendered wordmark width**, not a device category or component name. The React component observes its actual inline size when `opticalProfile="auto"`; callers may pin a profile for exports or controlled comparisons.

Profile	Rendered width	Treatment
Display	> 300px	Full luminous construction, ϕ annotations, canonical O network, and layered M.
Compact	121–300px	Stronger H stems, quiet continuous a/b crossbar, compact O network, and single compact M contour.
Micro	≤ 120px	Continuous H crossbar without ϕ ticks/brace/point, seven O chords without nodes, and a strengthened M contour.

Compact and micro assets apply small X-only spacing corrections recorded in the metrics contract. They never distort glyph geometry. The ϕ construction remains available in the display mark and explanatory animation, but it does not compete with letter recognition in utility sizes.

Perceptual validation protocol

The golden-ratio split remains an **unvalidated design prior**. Before making a preference claim, compare blinded display variants at 1:1, ϕ :1, and 2:1 while holding pillars, total crossbar span, stroke energy, and wordmark spacing constant. Randomize order and record THOM recognition, perceived balance, forced-choice preference, and response confidence.

Repeat the recognition check at 184px compact and 92px micro sizes. Those profiles intentionally suppress ratio annotations, so the small-size question is legibility and identity survival—not whether viewers can recover ϕ from the rendered pixels. Report sample, context, uncertainty, and null results alongside any winner.

Raster export examples

Target width	Uniform scale	Output height
460 px	1 px/u	120 px
920 px	2 px/u	240 px
1380 px	3 px/u	360 px
2300 px	5 px/u	600 px

SVG → Three.js translation

SVG's origin is at the upper left and y increases downward. The canonical Three.js scene uses the same 460u width and 120u height, but y increases upward.

```
type Point = { x: number; y: number };

const svgToThree = ({ x, y }: Point) =>
  new Vector3(x, 120 - y, 0);
```

Apply the function to every **M**, **L**, and **C** coordinate, including both cubic control points. Preserve **Z** as `closePath()`.

Preferred: flatten the SVG transform first

```
type SvgTransform = {
  tx: number;
  ty: number;
  sx: number;
  sy: number;
};

function transformedPoint({ x, y }: Point, t: SvgTransform) {
  const X = t.tx + t.sx * x;
  const Y = t.ty + t.sy * y;
  return new Vector3(X, 120 - Y, 0);
}
```

Flattening is least ambiguous because nested SVG transforms are resolved before the y-axis inversion.

Equivalent Three.js group transform

If local geometry has already been converted with `(x, 120 - y)` and the SVG transform is only translate + scale:

```
group.scale.set(sx, sy, 1);
group.position.set(tx, 120 - ty - 120 * sy, 0);
```

Camera

```
const camera = new OrthographicCamera(0, 460, 120, 0, -30, 30);
```

SVG datum	SVG y	Three.js y
Artboard top		0
Cap line		15
Construction axis		60
Baseline		104
Artboard bottom		120

Legacy 416 × 120 compatibility

The former 416 × 120 scene is a compatibility viewport, not the geometry master. If it cannot yet be migrated, preserve proportions with a uniform scale of **0.904347826**. The canonical art becomes **108.521739u** high with **5.739130u** of vertical inset above and below. Do not squeeze 460u into 416u while retaining the full 120u height.

Preservation and migration checklist

Preserve from the legacy implementation

- Metallic color ramps and light/dark/monochrome themes.
- Source-energy compensation and display stroke conversions.
- T rim, O perimeter/chord/node materials, H proportion construction, and M layered stroke materials.
- Animation timing, reveal ordering, H ϕ transition, O network stages, and M Fourier buildup.
- Responsive orthographic-camera behavior, adjusted to a 460×120 canonical base.

Replace during a future production migration

- Geometry and placement constants derived from the old 416×120 master.
- Legacy T path data and old per-glyph placements where they disagree with this canonical SVG.
- Any non-uniform whole-wordmark fit introduced solely to preserve the old viewport.

Acceptance checks

1. Composite `thom-canonical.svg` over `#c5b6f4` at 2300×600 and compare it to the refined audit raster.
2. Verify cap line, construction axis, baseline, and optical overshoots from the metrics JSON.
3. Confirm every cubic control point receives the same affine transform as its endpoint.
4. Confirm the 460×120 camera displays the canonical wordmark without cropping or anisotropic scaling.
5. Re-run the T grid measurements after any contour edit; do not copy old width tables forward.
6. Preserve color/material/motion sources until a separate migration explicitly supersedes them.
7. At widths of 92px, 184px, and 460px, verify the selected micro, compact, and display profiles respectively; every glyph must retain a high-contrast recognizable core.
8. Confirm compact and micro H output omits the ϕ point, ticks, and brace while the display H and ϕ -to-H animation retain them.
9. Do not describe the ϕ split as preferred until the blinded comparison protocol is run and reported; exact geometry alone is not preference evidence.

Specification artifacts: [canonical SVG](#) · [metrics JSON](#) · [print-ready PDF](#)